Do Two Notions of "Core" Agree? Core–Periphery Structure versus Structural Cohesion, Validated by Network Robustness

Anonymous Author(s)
Affiliation withheld for double-blind review

***Abstract*—The "core" of a network is defined in two largely separate ways: by density, as in core–periphery models and k-cores, and by resistance to disconnection, as in Moody and White's structural cohesion. We compare the two on eight networks from social, biological and infrastructure domains, using network robustness under targeted node removal as an external yardstick. Node by node, the notions almost coincide: k-coreness and cohesion level have Spearman correlation ≥ 0.99 in every network, and ordering nodes by cohesion dismantles networks no faster than ordering them by coreness. At the group level they diverge: in four of the eight networks a single connected k-core consists of many overlapping k-components (up to 7.7 times as many groups), a feature absent from all degree-preserving randomisations. We introduce *cohesive brokerage*, the number of extra cohesive groups a node belongs to, and show that removing brokers first beats both coreness and cohesion in all eight networks, lowering the robustness index by up to 44%; the gain grows with the group-level disagreement (Spearman 0.88). Rank hybrids of coreness and cohesion do not improve held-out performance. We also give an exact, certificate-accelerated algorithm for k-components and show that a widely used implementation can return non-maximal components.**

***Index Terms*—core–periphery structure, k-core, structural cohesion, k-components, node connectivity, network robustness, targeted attack**

I. Introduction

Almost every network analysis eventually asks which nodes form its "core". Two largely separate literatures answer this question differently. The *core–periphery* tradition, from Borgatti and Everett's ideal-pattern models [1] to continuous core scores [2] and the ubiquitous k-core decomposition [3], [4], defines the core through *density*: core nodes have many ties, especially to each other. The *structural cohesion* tradition of White, Harary, Moody and White [5], [6] defines it through *resistance to disconnection*: a group is cohesive to the extent that many nodes must be removed to pull it apart, which yields a nested hierarchy of k-components.

The two notions are related—every k-component lies inside the k-core—but not equivalent: two triangles sharing a node form a 2-core, yet removing the shared node disconnects them. How large is this difference in real networks, does it matter, and which notion should an analyst use? Despite the popularity of both ideas, they are rarely compared head-to-head, and we are not aware of a systematic comparison that also uses an external, functional yardstick.

We compare core–periphery cores (k-coreness, discrete Borgatti–Everett, Rombach et al.) with structural cohesion on eight networks from social, biological and infrastructure domains, and use network robustness under targeted node removal [7], [8] as the yardstick. We ask three questions. **RQ1:** How much do the two notions agree? **RQ2:** Which one identifies the nodes whose removal fragments the network fastest? **RQ3:** Does a hybrid of the two outperform either alone on networks it was not tuned on? We pre-stated the hypotheses that agreement is lower in spatial networks (H1), that cohesion-based orderings dismantle clustered networks faster than coreness (H2), and that a hybrid helps (H3).

Our main finding is that the two notions agree on *who* is central but disagree on *how* the centre is organised. Node by node, k-coreness and cohesion level are almost the same ranking (Spearman ≥ 0.99 in all eight networks), so H2 and H3 fail: cohesion levels add nothing to coreness for dismantling. At the level of groups, however, a single connected k-core often consists of many overlapping k-components, up to 7.7 times as many as there are k-core groups. This group-level gap is absent from degree-preserving randomisations and is tied together by a small set of *cohesive brokers*. Removing brokers first beats k-coreness and cohesion in every network, beats degree in four of eight, and reduces the robustness index by up to 44% relative to coreness, with gains that track the size of the group-level gap.

Contributions.

1) A node- and group-level comparison of core–periphery and cohesion cores across three domains, with simple propositions that characterise how the two can differ (Section III) and measures of group-level disagreement (fragmentation, group Jaccard).
2) Evidence from degree-preserving null models that the group-level gap is a genuine structural feature, consistent with the classical result that k-cores of random graphs are k-connected.
3) A functional validation showing that cohesion *levels* do not improve on coreness, whereas a new parameter-free score, *cohesive brokerage*, identifies critical nodes; an ablation shows that its signal comes from the higher levels of the cohesion hierarchy, not just articulation points.
4) A held-out evaluation showing that rank hybrids do not help, a negative result that guards against overfitting claims in this area.
5) An exact, certificate-accelerated k-component algorithm

with a correctness proof, which also exposes non-maximal output of the widely used NetworkX implementation; all code is released.

II. Related Work

Core–periphery structure. Borgatti and Everett [1] formalised the core–periphery (CP) ideal—a densely connected core and a periphery tied to the core but not to itself—and fitted it by maximising the correlation between the adjacency matrix and an ideal pattern, in discrete and continuous variants. Holme [9] proposed a closeness-based CP coefficient measured against degree-preserving randomisations. Rombach et al. [2], [10] replaced the binary partition by a family of transition profiles and aggregated over them to obtain a continuous core score. Later work detects multiple CP pairs [11] or fits generative block models [12]; Gallagher et al. [13] showed that "hub-and-spoke" and "layered" CP structures are distinct and that methods disagree on which they find. Csermely et al. [14] review the functional roles attributed to cores. A common thread is that these notions define the core by *density*.

k-cores. Seidman [3] introduced k-cores as a minimum-degree relaxation of cohesive subgroups; minimum degree k is necessary but not sufficient for k-connectivity. Core decomposition takes linear time [4], and coreness is a popular centrality: Kitsak et al. [15] found it predicts spreading influence better than degree, while Liu et al. [16] showed that densely knit "core-like groups" weakly attached to the rest of the network inflate coreness and mislead such rankings—a phenomenon closely related to the gap studied here. Burleson-Lesser et al. [17] linked k-core structure to the robustness of ecological and financial systems.

Structural cohesion. White and Harary [5] argued that cohesion should be measured by node connectivity rather than density, and Moody and White [6] defined the nested hierarchy of k-components and embeddedness. Exact computation is costly; Torrents and Ferraro [18] proposed heuristics that underlie the NetworkX implementation [19]. Connectivity testing relies on classical results of Even [20] and Esfahanian [21], and for random graphs Łuczak [22] showed that k-cores are k-connected with high probability.

Robustness and dismantling. Albert et al. [7] contrasted random failures with degree-targeted attacks; Schneider et al. [8] introduced the robustness index R; and optimal-percolation methods [23] provide state-of-the-art adaptive dismantling. Robustness gives an external, functional yardstick for comparing notions of "core", which we exploit.

Gap. CP models and k-cores on the one hand, and structural cohesion on the other, are rarely compared directly. We are not aware of a study that (i) quantifies their agreement at both the node and the group level across domains, (ii) tests the gap against a degree-preserving null model and (iii) asks which notion better identifies structurally critical nodes.

III. Preliminaries

Let $G = (V, E)$ be a simple, undirected, connected graph with $N = |V|$ nodes and $M = |E|$ edges, $N(v)$ the neighbourhood and $d_v = |N(v)|$ the degree of v, and $G[S]$ the subgraph induced by $S \subseteq V$.

Definition 1 (k-core, coreness [3])**.** *The k-core of G is the maximal induced subgraph with minimum degree at least k; it is obtained by repeatedly deleting nodes of degree $< k$. The* coreness c_v *is the largest k such that v belongs to the k-core. The* degeneracy *is $k_{\max} = \max_v c_v$.*

Definition 2 (Structural cohesion [5], [6])**.** *The node connectivity $\kappa(H)$ of a graph H is the minimum number of nodes whose removal disconnects H (for a complete graph, $|H|-1$). By Menger's theorem [24], $\kappa(H) \geq k$ iff every pair of nodes is joined by k internally node-disjoint paths. A* k-component *is a maximal set $K \subseteq V$ with $|K| > k$ and $\kappa(G[K]) \geq k$. The* cohesion level κ_v *is the largest k such that v lies in some k-component.*

k-components nest: every $(k+1)$-component lies inside a k-component, which yields Moody and White's hierarchy of embeddedness [6]. Unlike connected components, distinct k-components may overlap. We exploit that overlap explicitly:

Definition 3 (Cohesive brokerage)**.** *Let $m_k(v)$ be the number of k-components that contain v. The* cohesive brokerage *of v is $b_v = \sum_{k \geq 2} \max\{0,\, m_k(v) - 1\}$, i.e. the number of* additional *cohesive groups a node belongs to, summed over all levels of the hierarchy. Nodes with $b_v > 0$ are* brokers*.*

At $k = 2$, the 2-components are the biconnected components with at least three nodes and the brokers are the articulation points shared by them; for $k \geq 3$ brokerage generalises articulation to small vertex sets that glue together groups that cannot be merged into one k-cohesive group.

Proposition 1. *$\kappa_v \leq c_v$ for every node v.*

Proof. If K is a k-component, then $G[K]$ is k-connected with $|K| > k$, so every node of $G[K]$ has at least k neighbours in K (otherwise its neighbourhood, of size $< k$, would separate it). Hence K belongs to the k-core and $c_v \geq k$ for all $v \in K$. □

Proposition 2. *Every k-component lies inside a single connected component of the k-core. A connected component of the k-core that is itself k-connected is a k-component.*

Proof. The first claim follows from Proposition 1 and the connectivity of $G[K]$. For the second, let C be such a component and $K \supseteq C$ a k-component; by the first claim K lies in the component C of the k-core, so $K = C$. □

Proposition 3. *Two distinct k-components share at most $k-1$ nodes.*

Proof. Suppose $K_1 \neq K_2$ share $\geq k$ nodes and let S be any set of $< k$ nodes. Both $G[K_1] - S$ and $G[K_2] - S$ are connected and they share a node outside S, so $G[K_1 \cup K_2] - S$ is connected. Hence $K_1 \cup K_2$ is k-connected, contradicting the maximality of K_1 and K_2. □

Propositions 1–2 make precise *how* the two notions of "core" can differ: a node may sit deeper in the k-core hierarchy

than in the cohesion hierarchy (a node-level gap), and, even where the node levels coincide, one connected k-core can split into several overlapping k-components (a group-level gap). Proposition 3 shows that such splits are always glued together by fewer than k shared nodes, the brokers.

Example (bowtie). Take two triangles $\{A, B, X\}$ and $\{X, C, D\}$ sharing node X (Fig. 1a). Every node has degree ≥ 2, so the 2-core is the whole graph and $c_v = 2$ for all v. Removing X disconnects the graph, so $\kappa(G) = 1$, while each triangle is a 2-component. Thus $\kappa_v = c_v = 2$ for every node, but the single 2-core group consists of two 2-components overlapping in the broker X ($b_X = 1$). The node-level measures agree perfectly; the group-level structure does not, and the disagreement singles out the most critical node.

Random graphs. For Erdős–Rényi graphs the k-core, when it exists, is k-connected with high probability [22]. Hence, in a random graph, both node-level and group-level gaps vanish asymptotically, which makes degree-preserving randomisation a natural null model for both gaps (Section IV-D).

IV. Methodology

A. Core–periphery scores

k-core coreness. We use the $O(M)$ peeling algorithm of Batagelj and Zaveršnik [4].

Discrete Borgatti–Everett (BE). Each node is labelled core ($x_v = 1$) or periphery ($x_v = 0$). The ideal pattern is $\delta_{uv} = 1$ unless both u and v are periphery, and the fit is the Pearson correlation ρ between $\{A_{uv}\}_{u<v}$ and $\{\delta_{uv}\}_{u<v}$ [1]. With $\Pi = N(N-1)/2$ node pairs, n_p periphery nodes, E_{pp} periphery–periphery edges, $P = \Pi - n_p(n_p-1)/2$ and $\Sigma = M - E_{pp}$,

$$\rho = \frac{\Pi\,\Sigma - MP}{\sqrt{(\Pi M - M^2)(\Pi P - P^2)}}. \tag{1}$$

Flipping one node changes n_p by one and E_{pp} by its number of periphery neighbours, so the effect of a flip is evaluated in $O(1)$ and applied in $O(d_v)$. We run greedy label switching: starting from the top 10% of nodes by degree as core, visit nodes in random order, flip whenever ρ increases and stop after a pass without improvement; we keep the best of 20 random visiting orders.

Rombach et al. Nodes are assigned the values of the profile [2, Eq. 9]

$$C^*_i = \begin{cases} \dfrac{i(1-\alpha)}{2\lfloor \beta N \rfloor}, & i \leq \lfloor \beta N \rfloor, \\ \dfrac{(i - \lfloor \beta N \rfloor)(1-\alpha)}{2(N - \lfloor \beta N \rfloor)} + \dfrac{1+\alpha}{2}, & i > \lfloor \beta N \rfloor, \end{cases} \tag{2}$$

normalised to unit sum, and the assignment (a permutation) maximising the core quality $R_{\alpha,\beta} = \sum_{u,v} A_{uv} C_u C_v$ is sought. The aggregate core score is

$$CS(v) = Z \sum_{\alpha,\beta} C_v(\alpha,\beta)\, R_{\alpha,\beta}, \tag{3}$$

with Z such that $\max_v CS(v) = 1$. We use $\alpha \in \{0, 0.2, \ldots, 1\}$ and $\beta \in \{0.1, 0.2, \ldots, 0.9\}$ (54 pairs; the original paper uses a 100×100 grid, which is computationally prohibitive for our larger networks). For each pair we run simulated annealing [25] over swaps of two nodes' values, started from the degree ordering, with $\max(500N, 3\times10^5)$ proposals and geometric cooling from T_0 (the mean $|\Delta R|$ of 2000 random proposals) to $10^{-6}T_0$. The change in R of a swap is $\Delta R = 2\Delta_u(s_u - s_v) - 2A_{uv}\Delta_u^2$ with $s = AC$ and $\Delta_u = C_v - C_u$, so each proposal costs $O(\log d_u)$ and each accepted move $O(d_u + d_v)$.

Algorithm 1 Exact k-components by certified cut splitting

```
K_2 ← biconnected components of G with ≥ 3 nodes [26]
stack ← {(K, 3) : K ∈ K_2}
while stack not empty do
    (P, k) ← pop; H ← k-core of G[P]
    for each connected component C of H with |C| > k do
        S ← SmallCut(G[C], k)
        if S = ∅ then
            add C to K_k; push (C, k+1)
        else
            push (D ∪ S, k) for each component D of G[C] − S
        end if
    end for
end while
function SmallCut(H, k)                    ▷ a cut of size < k, or ∅
    v ← a minimum-degree node of H
    for each pair (x, y) ∈ P_v do          ▷ see text
        if not Cert(x, y, k) and κ(x, y) < k then
            return a minimum x–y vertex cut
        end if
    end for
    return ∅
end function
```

B. Exact structural cohesion

NetworkX's `k_components` [19], which follows the heuristics of Torrents and Ferraro [18], is the standard tool for this task. We found that it can return *non-maximal* components (e.g. for $k = 3$ in the Les Misérables graph it reports a 32-node set that is strictly contained in a 3-connected 34-node set; see also Table VIII). We therefore implement an exact algorithm (Algorithm 1) and verify it against exhaustive enumeration.

Correctness. *(i) Splitting is safe.* If S is a vertex cut of $G[C]$ with $|S| < k$ and K is a k-component inside C, then $G[K]-S$ is connected, so $K \setminus S$ lies within one component D of $G[C]-S$ and $K \subseteq D \cup S$. Pruning to the k-core never removes a node of K (Proposition 1). Hence no k-component is lost, and a piece that is k-connected equals the k-component containing it, i.e. it is maximal. The recursion terminates because each piece $D \cup S$ is a proper subset of C. *(ii) The test is complete.* With v of minimum degree, $G[C]$ is k-connected iff $\kappa(x, y) \geq k$ for every pair in $\mathcal{P}_v = \{(v, w) : w \notin N(v) \cup \{v\}\} \cup \{(x, y) : x, y \in N(v),\, xy \notin E\}$, i.e. for v and each of its non-neighbours and for each non-adjacent pair of its neighbours [20], [21].

Certificates. Most pairs (92% in our networks, Section VI) are certified without a max-flow computation:

Lemma 1. *Let T_x be the smallest set that contains x and $N(x)$, contains every node verified by max-flow to satisfy $\kappa(x, \cdot) \geq k$, and contains every node with at least k neigh-*

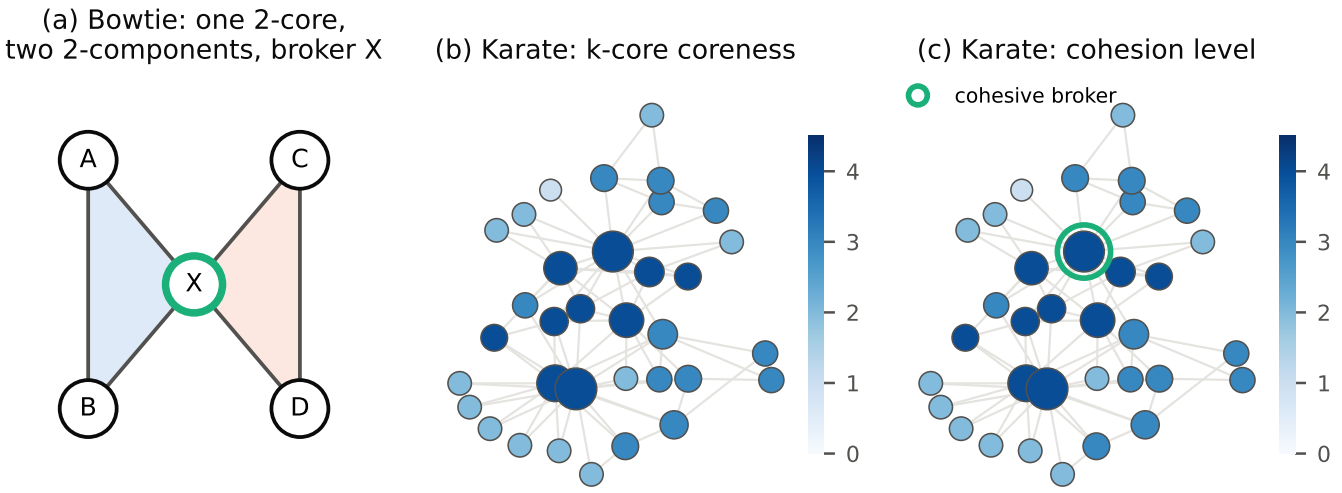


Fig. 1. (a) The bowtie: a single 2-core that consists of two overlapping 2-components; their overlap X is a cohesive broker. (b, c) Zachary's karate club coloured by k-core coreness and by cohesion level. The two scores coincide at every node, yet the cohesion hierarchy identifies a broker (ringed) between the two 2- and 3-components.

bours in T_x ("bootstrap closure"). Then $\kappa(x, y) \geq k$ for every $y \in T_x$. Moreover, $\kappa(x, y) \geq k$ whenever $|N(x) \cap N(y)| \geq k$.

Proof. The second claim holds because common neighbours give k internally disjoint paths of length two. For the first, let S be any vertex set with $|S| < k$ and $x \notin S$. We show by induction over the order of insertion that every $w \in T_x \setminus S$ is connected to x in $G - S$. This holds for x, for neighbours of x and for flow-verified nodes. A node added by the closure has at least $k > |S|$ neighbours in T_x, so at least one of them, inserted earlier, lies outside S and is connected to x by the induction hypothesis. □

Remaining pairs are decided by a unit-capacity augmenting-path max-flow on the node-split graph with cutoff k, compiled with Numba [27]; a minimum vertex cut is read off the residual graph.

C. Agreement measures (RQ1)

Node level. For each pair of score vectors we report Spearman's ρ_s; the Jaccard overlap of the top-$x\%$ sets ($x \in \{5, 10, 20\}$), where a top set contains every node scoring at least the $\lceil xN \rceil$-th largest value (so tied nodes are included together and identical rankings give overlap 1; the lowest tier, e.g. the BE periphery, is never part of a top set); and the adjusted mutual information (AMI) [28] between binary core labels (main k-core; deepest k-components; the BE labelling; the top-|BE core| nodes by CS). **Group level.** For every $k \geq 2$ we compare the connected components of the k-core ($n_{cc}(k)$ groups) with the k-components ($n_{kc}(k)$ groups) through the *fragmentation* $F = \sum_k n_{kc}(k) / \sum_k n_{cc}(k)$ (by Proposition 2, $F = 1$ when every k-core group is k-cohesive), the *coverage* (fraction of k-core nodes that lie in some k-component) and the *group Jaccard* J_G: the size-weighted best-match Jaccard similarity of each k-core group to a k-component. Coverage and J_G are averaged over k with weights equal to the k-core size.

D. Null model

For each network we generate 100 degree-preserving randomisations by $10M$ accepted double-edge swaps [29] (rejecting self-loops and multi-edges), keep the largest connected component, and recompute all statistics (BE uses 5 restarts). We report z-scores and empirical two-sided p-values $p = \min\{1, 2\min(p_{\geq}, p_{\leq})\}$ with $p_{\geq} = (1 + \#\{\text{null} \geq \text{obs}\})/101$.

E. Robustness experiment (RQ2)

Nodes are removed in a given order; $S(q)$ is the size of the largest connected component, relative to N, after removing a fraction q of nodes, and $R = \frac{1}{N}\sum_{Q=1}^{N} S(Q/N)$ [8]. A lower R means the ordering found more critical nodes. Curves are computed exactly in $O(M\,\alpha(N))$ time by re-inserting nodes in reverse order with union–find [30]. We compare static orderings by degree, betweenness (exact, Brandes [31]), k-core coreness, BE labels, Rombach CS, cohesion level and cohesive brokerage, against the mean of 30 random orderings and an *adaptive* degree attack (recomputing degrees after every removal) [7], a strong and widely used baseline. Coreness, cohesion and brokerage take few distinct values, so ties matter: our main analysis breaks ties by degree (then node index); as a sensitivity analysis we break all ties uniformly at random and average R over 50 draws.

F. Hybrid scores (RQ3)

With $r(\cdot)$ the rank of a score normalised to $[0, 1]$, we define $H_v(\lambda) = \lambda\, r(c_v) + (1 - \lambda)\, r(x_v)$, with $x = \kappa$ (*core+cohesion*, as originally planned) or $x = b$ (*core+brokerage*), ties broken by degree. The weight $\lambda \in \{0, 0.1, \ldots, 1\}$ is chosen on training networks by minimising the mean of $R(\lambda)/R_{\text{k-core}}$

TABLE I

NETWORK STATISTICS AFTER PREPROCESSING (LARGEST CONNECTED COMPONENT). C: AVERAGE CLUSTERING; r: DEGREE ASSORTATIVITY; $k_{\max}$: DEGENERACY; $\kappa_{\max}$: MAXIMUM COHESION LEVEL.

Network	Dom.	N	M	$\langle d \rangle$	C	r	$k_{\max}$	$\kappa_{\max}$
Karate	Soc.	34	78	4.6	0.57	-0.48	4	4
GrQc	Soc.	4,158	13,422	6.5	0.56	0.64	43	43
Email-EU	Soc.	986	16,064	32.6	0.41	-0.03	34	34
Facebook	Soc.	4,039	88,234	43.7	0.61	0.06	115	115
Yeast PPI	Bio.	1,004	8,319	16.6	0.65	0.56	40	40
C. elegans	Bio.	297	2,148	14.5	0.29	-0.16	10	10
Power grid	Inf.	4,941	6,594	2.7	0.08	0.00	5	5
Air India	Inf.	68	179	5.3	0.61	-0.48	6	6

(ties resolved towards larger λ) and evaluated only on held-out networks, using (i) a domain-stratified 2-fold split (fold A: Karate, Email-EU, Yeast, Power grid; fold B: GrQc, Facebook, C. elegans, Air India), applied in both directions, and (ii) leave-one-network-out.

G. Statistical comparison

With eight networks we report, for each pair of orderings, the mean difference in R with a 95% bootstrap confidence interval (10^4 resamples of networks [32]), win counts, the exact two-sided Wilcoxon signed-rank test [33] and the matched-pairs rank-biserial correlation r_{rb} as effect size [34]; p-values are Holm-adjusted within each family of comparisons [35]. With $n = 8$ the smallest attainable exact two-sided p is $2/2^8 \approx 0.008$, so we emphasise effect sizes over significance.

V. DATASETS

We study eight networks from three domains (Table I). *Social:* Zachary's karate club [36]; the arXiv General Relativity co-authorship network (GrQc) [37]; the e-mail network of a European research institution (Email-EU) [37], [38]; and the combined Facebook ego networks [39], all from SNAP [40]. *Biological:* the yeast protein–protein interaction network of Collins et al. [41] and the *C. elegans* neural network [42], [43], both from Netzschleuder [44]. *Infrastructure:* the Western US power grid [43] (Netzschleuder) and the Indian domestic air-route network, built from OpenFlights [45] by keeping routes whose origin and destination airports are both in India. Every network is converted to a simple undirected unweighted graph (directions, weights, self-loops and multi-edges are dropped) and reduced to its largest connected component. The sample deliberately spans sparse, low-clustering spatial networks (power grid) to dense, highly clustered social networks (Facebook, GrQc).

VI. RESULTS

A. RQ1: Do the two notions of core agree?

Node level: almost perfectly for k-cores, weakly for CP models. Table II and Figs. 2–3 summarise agreement with the cohesion level κ. k-core coreness and cohesion are nearly interchangeable as rankings: Spearman $\rho_s \geq 0.99$ and top-10% Jaccard $J_{10} \geq 0.96$ in every network, and the main

TABLE II

AGREEMENT WITH STRUCTURAL COHESION. NODE LEVEL: SPEARMAN ρ_s OF κ WITH k-CORENESS c, BE LABELS AND ROMBACH CS; JACCARD OF THE (TIE-INCLUSIVE) TOP-10% SETS OF c AND κ; PERCENTAGE OF GAP NODES ($c_v > \kappa_v$). GROUP LEVEL: FRAGMENTATION F, GROUP JACCARD J_G, COVERAGE AND NUMBER OF COHESIVE BROKERS.

	Node level					Group level			
	ρ_s with κ			J_{10}	gap				
Network	c	BE	CS	c,κ	%	F	J_G	cov.	brok.
Karate	1.00	0.54	0.78	1.00	0.0	1.67	0.86	1.00	1
GrQc	0.99	0.37	0.92	1.00	4.6	5.72	0.71	0.99	558
Email-EU	1.00	0.40	0.99	1.00	0.4	1.00	1.00	1.00	0
Facebook	1.00	0.39	0.98	0.96	15.7	4.66	0.53	0.99	101
Yeast PPI	1.00	0.36	0.95	1.00	8.3	4.25	0.63	0.99	80
C. elegans	1.00	0.24	0.95	1.00	0.0	1.00	1.00	1.00	0
Power grid	0.99	0.27	0.73	1.00	1.2	7.67	0.88	0.98	71
Air India	1.00	0.36	0.98	1.00	0.0	1.60	0.92	1.00	4

core coincides exactly with the deepest k-components (AMI $= 1$) in all eight. The fraction of *gap nodes* ($c_v > \kappa_v$) is zero in four networks and at most 15.7% (Facebook); because 87% of gap nodes lose a single level, the cohesion deficit $1-\sum_v \kappa_v / \sum_v c_v$ never exceeds 1%. The classical CP models agree much less: Rombach's CS reaches $\rho_s = 0.73$–0.99 and BE's binary labels only $\rho_s = 0.24$–0.54. Across domains, agreement of the CP models with cohesion is lowest for the infrastructure networks (mean ρ_s for CS: social 0.92, biological 0.95, infrastructure 0.85), which is the pattern anticipated by H1 for spatial networks; for k-coreness, however, agreement is uniformly near perfect (0.994–0.999), so H1 is *not* supported for the most widely used CP proxy.

Group level: substantial disagreement in half of the networks. Node levels hide how the cores are organised. In Email-EU and *C. elegans* every connected k-core group is k-cohesive at every k ($F = 1$, $J_G = 1$): the two notions of core are identical. In GrQc, Facebook, Yeast and the power grid, by contrast, a single connected k-core repeatedly breaks into many overlapping k-components ($F = 4.2$–7.7; Fig. 4). GrQc's 2-core is one connected group but contains 260 2-components; on Facebook, k-core groups overlap their best-matching k-component with a size-weighted Jaccard of only $J_G = 0.53$. The cores are therefore *the same nodes, organised differently*: coverage stays ≥ 0.98, so almost every k-core node does belong to some k-component, but the k-core glues several cohesive groups together through a small number of brokers (71–558 nodes, 1–13% of the network).

What drives the gap? Across the eight networks, the size-normalised group disagreement $1 - J_G$ increases with transitivity ($\rho_s = 0.67$) and degree assortativity (0.64), but not with size (0.40), and the share of gap nodes is strongly associated with assortativity ($\rho_s = 0.88$, $p = 0.004$). The raw fragmentation F, in contrast, grows with network size ($\rho_s = 0.78$), because larger networks simply contain more groups; we therefore treat F as descriptive and rely on the null model to separate structure from size.

The gap is not an artefact of the degree sequence.

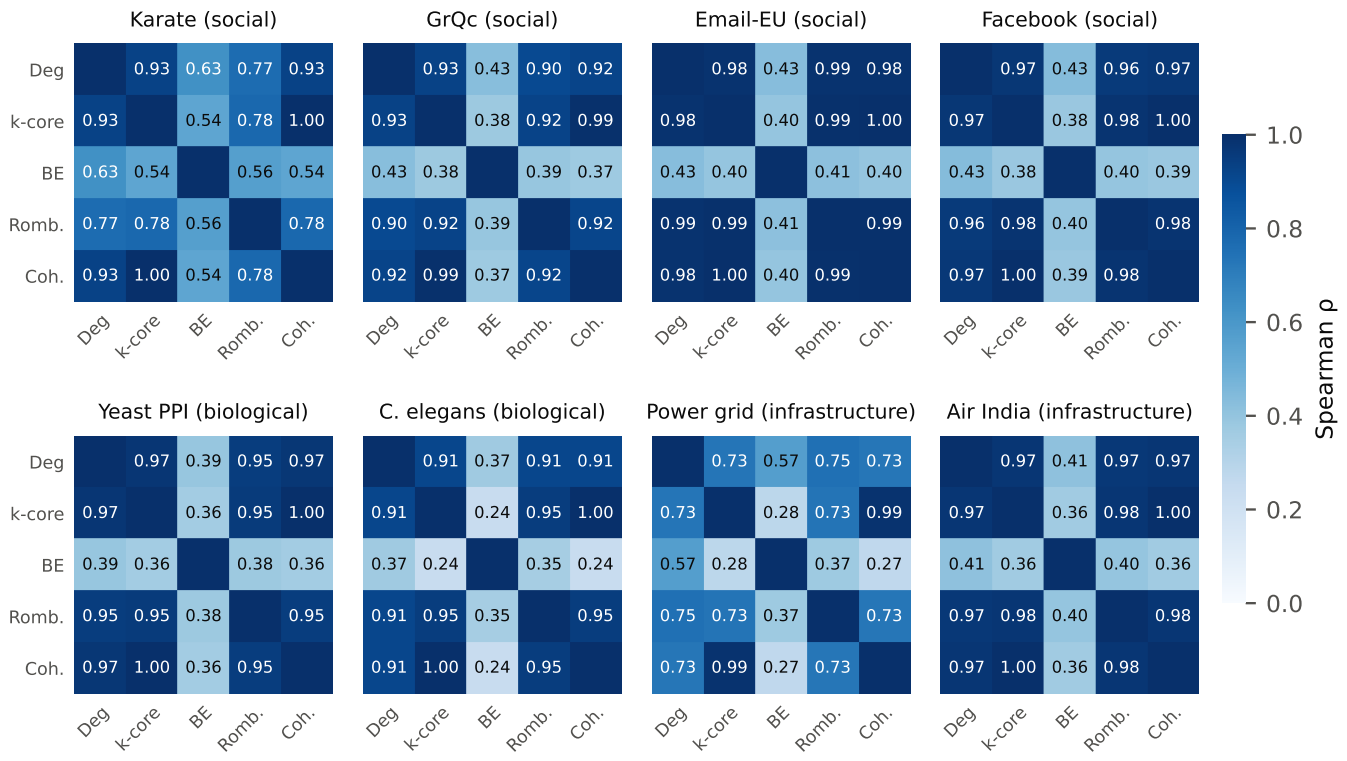


Fig. 2. Node-level Spearman correlations between degree, k-coreness, BE labels, Rombach CS and cohesion level in each network.

In degree-preserving randomisations of the same networks the gap vanishes (Table III, Fig. 5): randomised graphs have essentially no gap nodes, $F \approx 1$ and no brokers, as predicted by Łuczak's theorem for random graphs [22]. For GrQc, Facebook, Yeast and the power grid the observed fragmentation and broker fraction lie above all 100 randomisations (empirical $p = 0.02$, the smallest attainable value). Only in the two smallest networks (Karate, Air India) do randomisations occasionally produce brokers, a finite-size effect. Interestingly, the BE correlation of five of the eight empirical networks is significantly *lower* than that of their randomisations (e.g. $z = -83$ for Facebook, -54 for GrQc and -48 for Yeast; $p = 0.02$ in all five): clustered networks are less core–periphery-like, in the BE sense, than random graphs with the same degrees—plausibly because rewiring replaces local, clustered ties by ties among high-degree nodes, which the BE pattern rewards. Of the two notions, it is the cohesion structure—not the density-based CP fit—that carries non-random signal in these networks.

TABLE III
OBSERVED STATISTICS VERSUS THE MEAN OF 100 DEGREE-PRESERVING RANDOMISATIONS, AND THE z-SCORE OF THE BE CORRELATION ρ^*.

Network	gap (%) obs.	gap (%) null	F obs.	F null	brokers (%) obs.	brokers (%) null	BE ρ^* z
Karate	0.0	0.59	1.67	1.05	2.9	0.65	2.7
GrQc	4.6	0.00	5.72	1.01	13.4	0.00	-54.0
Email-EU	0.4	0.00	1.00	1.00	0.0	0.00	-9.7
Facebook	15.7	0.00	4.66	1.00	2.5	0.00	-83.3
Yeast PPI	8.3	0.00	4.25	1.00	8.0	0.00	-47.7
C. elegans	0.0	0.00	1.00	1.00	0.0	0.00	0.4
Power grid	1.2	0.00	7.67	1.12	1.4	0.00	-2.8
Air India	0.0	0.00	1.60	1.04	5.9	0.40	-0.4

B. RQ2: Which notion finds the critical nodes?

Table IV lists the robustness index of every ordering, Fig. 6 shows the curves, and Table V the paired comparisons.

Cohesion level does not beat coreness. Because the two rankings are almost identical, so are their attacks: the mean difference is $\overline{\Delta R} = +0.002$ (95% CI $[-0.001, 0.005]$; 2 wins, 3 losses, 3 exact ties). H2 is therefore not supported. Both are clearly *worse* than plain degree ($\overline{\Delta R} = +0.029$ for coreness, $0/8$ wins, $r_{rb} = 1$), confirming that deep k-shells are not where a network is most vulnerable [16]. Rombach's CS is worse still on average, while BE's "core first, then by degree" ordering is statistically indistinguishable from degree.

Cohesive brokerage does. Promoting brokers to the front of the degree ordering beats both k-coreness and cohesion in all eight networks ($\overline{\Delta R} = -0.049$, CI $[-0.086, -0.017]$, $r_{rb} = -1$), and beats or ties degree in six of eight ($\overline{\Delta R} = -0.020$, CI $[-0.047, 0.000]$). The gains are concentrated where the group-level gap is large: relative to degree, brokerage lowers R by 24% on Facebook and 33% on Yeast, and relative to k-coreness by 33–44% on Facebook, GrQc and Yeast; on Email-EU and *C. elegans*, which have no brokers, it reduces

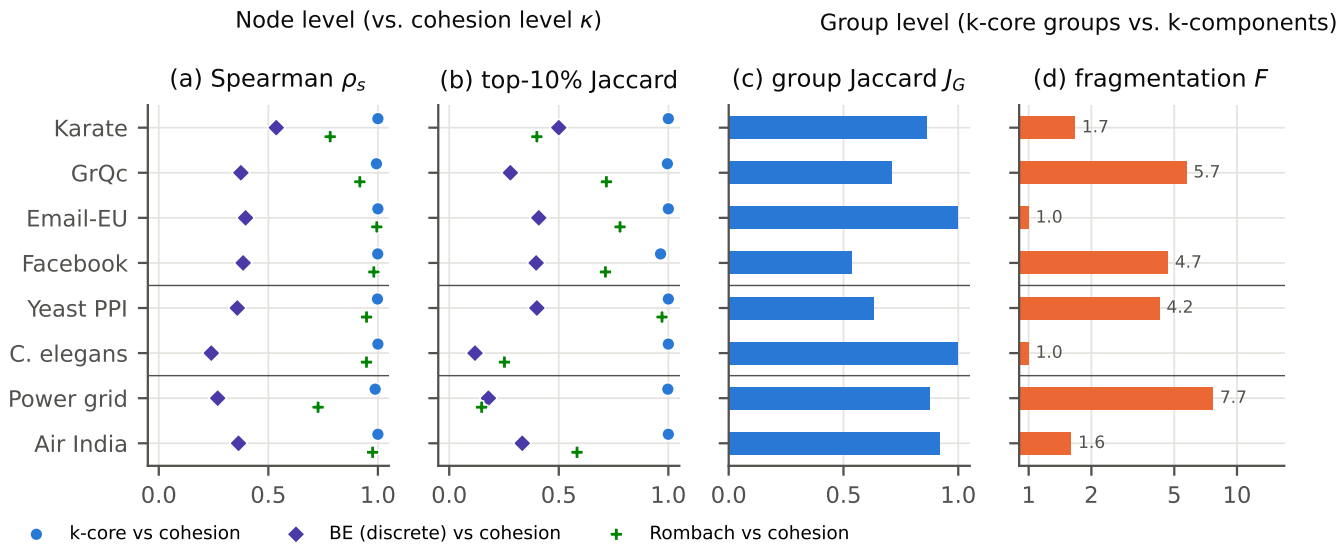


Fig. 3. Node-level agreement (a, b) is near perfect between k-coreness and cohesion in every network, whereas group-level agreement (c, d) varies widely: k-core groups split into up to 7.7 times as many k-components.

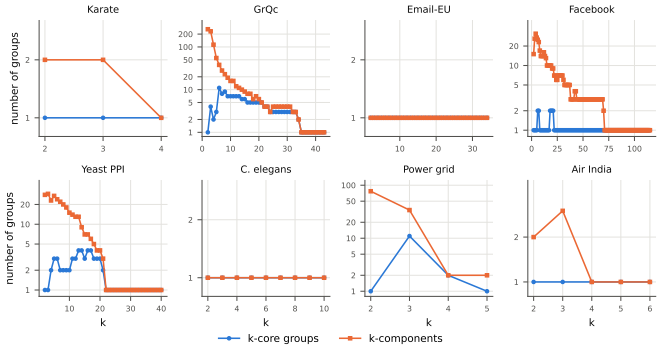


Fig. 4. Number of connected k-core groups and of k-components for every k (log scale). In Email-EU and *C. elegans* the two curves coincide.

to degree. Across networks, the advantage of brokerage over k-coreness tracks the group disagreement $1 - J_G$ ($\rho_s = 0.88$, $p = 0.004$), and so does its advantage over degree ($\rho_s = 0.75$, $p = 0.03$), whereas the latter is unrelated to size ($\rho_s = 0.08$). Brokerage is not a universal winner: betweenness, which measures brokerage along shortest paths, remains the strongest static ordering on GrQc, Email-EU, Facebook and Yeast, and an adaptive degree attack is stronger than brokerage in six of eight networks. Among static orderings, brokerage's mean rank (2.44) is comparable to BE (2.25), betweenness (2.75) and degree (2.81), and far ahead of k-coreness (5.56) and cohesion (5.81); unlike betweenness it requires no shortest-path computation once the cohesion hierarchy is known. With eight networks and Holm correction over twelve comparisons, no adjusted p-value falls below 0.094 (the minimum attainable is ≈ 0.09); the conclusions rest on the consistent direction and the effect sizes.

Brokerage is more than articulation. Table VI restricts brokerage to $k = 2$ (articulation points shared by 2-components) or to $k \geq 3$, and compares it with all articulation points. Classical articulation points help in some networks but are harmful in the power grid ($R = 0.113$ versus 0.063 for degree), where most of them only cut off small tree-like branches. On Facebook the entire gain comes from higher-order brokers ($k \geq 3$: $R = 0.224$ versus 0.283 for both articulation variants), and on Yeast only the full hierarchy achieves the best result (0.185). The multi-level cohesion hierarchy, not just biconnectivity, carries the signal.

Sensitivity to tie-breaking. When ties are broken uniformly at random (Table V, right; Fig. 7b), every coarse score deteriorates, and brokerage most of all, since 86–100% of nodes share the value $b = 0$. Brokerage is thus a sparse *prioritisation layer*: it tells which nodes to remove first, and must be combined with a base ranking for the rest. Coreness and cohesion likewise depend on their tie-breaker: their advantage over random removal shrinks in every network (e.g. in the power grid from 0.129 to 0.055 for coreness), which underlines that tie-breaking rules must be reported in robustness studies.

C. RQ3: Does a hybrid help?

Neither hybrid improves on its better component on held-out networks (Table VII, Fig. 8). For core+cohesion the two inputs are nearly the same ranking; intermediate weights mix two degree tie-breaks and are slightly worse ($\approx 1\%$) than either extreme, the selected λ^* flips between 0 and 1 across folds, and the held-out hybrid never wins (H3 not supported). For core+brokerage the training folds always select $\lambda^* = 0$, i.e. pure brokerage: adding coreness to brokerage only hurts (Fig. 8b), so the best "hybrid" is brokerage itself, which matches the conclusion of RQ2.

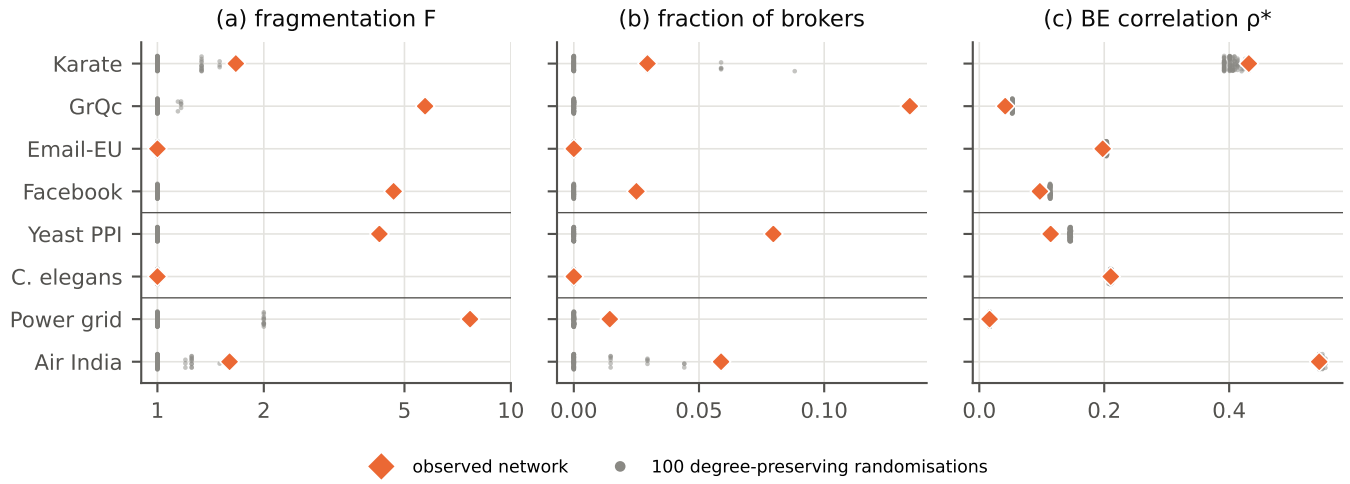


Fig. 5. Observed values (diamonds) against 100 degree-preserving randomisations (grey) for fragmentation F (log scale), the fraction of cohesive brokers and the BE correlation ρ^*.

TABLE IV

ROBUSTNESS INDEX R (LOWER IS BETTER) FOR EACH STATIC ORDERING; TIES BROKEN BY DEGREE. BOLD: BEST STATIC ORDERING; DEG.†: ADAPTIVE DEGREE ATTACK, UNDERLINED WHERE IT BEATS EVERY STATIC ORDERING. BOTTOM ROW: MEAN RANK AMONG THE SEVEN STATIC ORDERINGS.

Network	Rand.	Deg.	Betw.	c	BE	CS	κ	b	Deg.†
Karate	0.404	0.142	0.145	0.155	0.142	0.237	0.155	**0.137**	0.136
GrQc	0.368	0.131	**0.086**	0.198	0.127	0.190	0.197	0.128	0.107
Email-EU	0.474	0.315	**0.293**	0.327	0.315	0.328	0.327	0.315	0.292
Facebook	0.456	0.295	**0.104**	0.336	0.295	0.342	0.343	0.224	0.266
Yeast PPI	0.382	0.274	**0.129**	0.327	0.274	0.321	0.336	0.185	0.218
C. elegans	0.477	**0.331**	0.348	0.344	**0.331**	0.383	0.344	**0.331**	0.286
Power grid	0.213	0.063	0.066	0.084	**0.059**	0.165	0.081	0.065	0.052
Air India	0.408	**0.070**	0.079	0.080	**0.070**	0.092	0.080	0.074	0.064
Mean rank	–	2.81	2.75	5.56	2.25	6.38	5.81	2.44	–

TABLE V

PAIRED COMPARISONS OVER THE EIGHT NETWORKS. $\overline{\Delta R} = \overline{R_A - R_B}$ (NEGATIVE: A DISMANTLES FASTER), BOOTSTRAP 95% CI, WINS/LOSSES OF A, HOLM-ADJUSTED EXACT WILCOXON p AND RANK-BISERIAL r_{rb}; RIGHT: THE SAME COMPARISON WHEN ALL TIES ARE BROKEN AT RANDOM.

	ties broken by degree					random ties	
A vs. B	$\overline{\Delta R}$	95% CI	W/L	p_{Holm}	r_{rb}	$\overline{\Delta R}$	r_{rb}
κ vs. c	+0.002	[-0.001, +0.005]	2/3	1.000	+0.33	+0.005	+0.72
κ vs. BE	+0.031	[+0.017, +0.048]	0/8	0.094	+1.00	-0.023	-0.39
κ vs. CS	-0.024	[-0.051, -0.000]	5/3	1.000	-0.50	+0.015	+0.61
κ vs. Deg.	+0.030	[+0.017, +0.048]	0/8	0.094	+1.00	+0.069	+1.00
b vs. c	-0.049	[-0.086, -0.017]	8/0	0.094	-1.00	+0.036	+0.44
b vs. κ	-0.051	[-0.090, -0.018]	8/0	0.094	-1.00	+0.031	+0.44
b vs. Deg.	-0.020	[-0.047, +0.000]	4/2	1.000	-0.62	+0.100	+0.83
b vs. Deg.†	+0.005	[-0.015, +0.022]	2/6	1.000	+0.28	+0.126	+0.94
b vs. Betw.	+0.026	[-0.000, +0.059]	4/4	1.000	+0.44	+0.147	+1.00
c vs. Deg.	+0.029	[+0.015, +0.044]	0/8	0.094	+1.00	+0.064	+1.00
CS vs. c	+0.026	[+0.004, +0.051]	2/6	0.656	+0.67	-0.010	-0.28
BE vs. c	-0.030	[-0.046, -0.016]	8/0	0.094	-1.00	+0.027	+0.44

TABLE VI

BROKERAGE ABLATION: NUMBER OF NODES FLAGGED AND R WHEN THEY ARE REMOVED FIRST (THEN BY DEGREE): ALL ARTICULATION POINTS, BROKERS AT $k = 2$ ONLY, AT $k \geq 3$ ONLY, AND AT ALL LEVELS (b).

	# nodes			R (ties broken by degree)				
Network	art.	$b_{k=2}$	$b_{k\geq3}$	Deg.	art.	$b_{k=2}$	$b_{k\geq3}$	b
Karate	1	1	1	0.142	**0.137**	**0.137**	**0.137**	**0.137**
GrQc	701	198	434	0.131	0.135	**0.126**	0.128	0.128
Email-EU	73	0	0	0.315	**0.304**	0.315	0.315	0.315
Facebook	11	7	101	0.295	0.283	0.283	0.224	**0.224**
Yeast PPI	124	18	68	0.274	0.198	0.230	0.240	**0.185**
C. elegans	3	0	0	0.331	**0.330**	0.331	0.331	0.331
Power grid	1229	58	15	0.063	0.113	0.065	**0.063**	0.065
Air India	6	1	4	**0.070**	0.071	**0.070**	0.074	0.074

D. Computation

Table VIII compares our exact algorithm with NetworkX on the same machine (one process, after a warm-up run that excludes Numba's one-off compilation). Our algorithm computes the full k-component hierarchy of every network

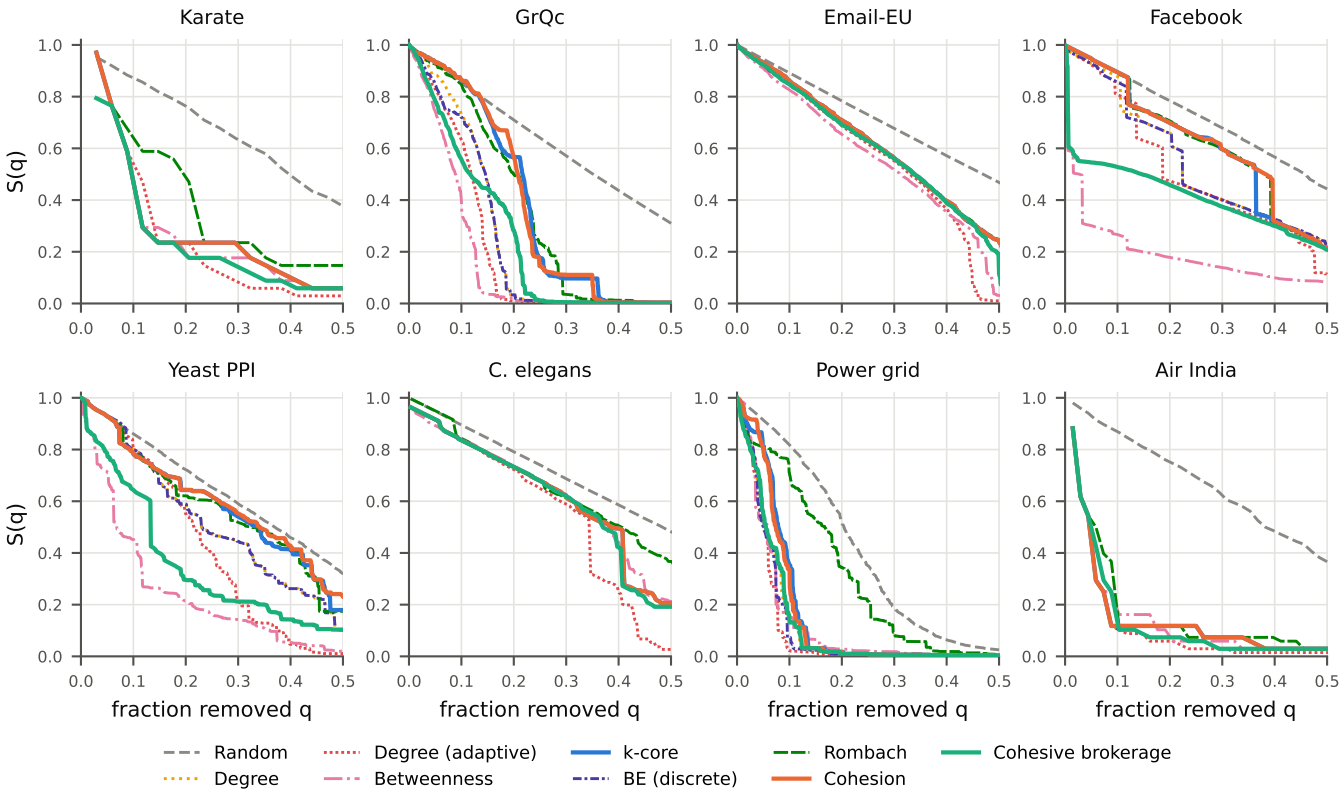


Fig. 6. Robustness curves $S(q)$ for all orderings (first half of the removal process).

TABLE VII
HELD-OUT EVALUATION OF THE HYBRIDS. λ^*: SELECTED WEIGHTS; MEAN R OF THE HYBRID, OF k-CORENESS AND OF THE PARTNER SCORE $x \in \{\kappa, b\}$ OVER THE TEST NETWORKS; WINS/LOSSES AGAINST THE BETTER SINGLE SCORE.

Hybrid	scheme	λ^*	$\overline{R}_H$	$\overline{R}_c$	$\overline{R}_x$	W/L vs. best
$c{+}\kappa$	2-fold	0.0, 1.0	0.232	0.231	0.233	0/2
$c{+}\kappa$	leave-one-out	0.0, 1.0	0.233	0.231	0.233	0/4
$c{+}b$	2-fold	0.0	0.182	0.231	0.182	0/0
$c{+}b$	leave-one-out	0.0	0.182	0.231	0.182	0/0

TABLE VIII
RUNTIME OF THE EXACT k-COMPONENT COMPUTATION (OURS) AND OF THE NETWORKX k-COMPONENT ROUTINE (TIME LIMIT 10 MIN), NUMBER OF LEVELS k AT WHICH THE OUTPUTS DIFFER, AND NUMBER OF NETWORKX COMPONENTS THAT ARE STRICTLY CONTAINED IN ONE OF OURS (I.E. NOT MAXIMAL).

Network	ours (s)	NetworkX (s)	levels differ	non-max. sets
Karate	0.003	0.18	0	0
GrQc	30.2	>600	–	–
Email-EU	4.3	>600	–	–
Facebook	83.4	>600	–	–
Yeast PPI	4.5	>600	–	–
C. elegans	0.198	>600	–	–
Power grid	0.113	>600	–	–
Air India	0.020	0.66	3	4

in at most 84 s (Facebook, $k_{\max} = 115$), and of the power grid and *C. elegans* in under 0.25 s; NetworkX finishes only the two smallest networks within the 10-minute limit, and on Air India it returns four non-maximal components at three levels of the hierarchy. The certificates of Lemma 1 are what make this possible: across all networks, 92% of the node pairs examined by the connectivity test (62–96% per network) are certified without any max-flow computation.

VII. Discussion and Limitations

Two notions, one ranking, two organisations. Proposition 1 guarantees $\kappa_v \le c_v$, and our data show that the inequality is almost always tight: real k-cores are almost k-cohesive *node by node*. The density view and the connectivity view therefore rank nodes nearly identically, and for tasks that only need a ranking—for example choosing seeds or computing a centrality—the linear-time k-core is an excellent proxy for the far more expensive cohesion level. The two views part ways at the level of *groups*. In highly clustered networks (GrQc, Facebook, Yeast) and in the sparse power grid, k-cores are federations of cohesive groups joined through a few shared nodes; in Email-EU and *C. elegans*, which are dense and well mixed, they are not. This matches the intuition behind "core-like groups" [16] and structural holes [46]: density aggregates groups that connectivity keeps apart. It also explains why the CP models agree less with cohesion: they are tuned to a single

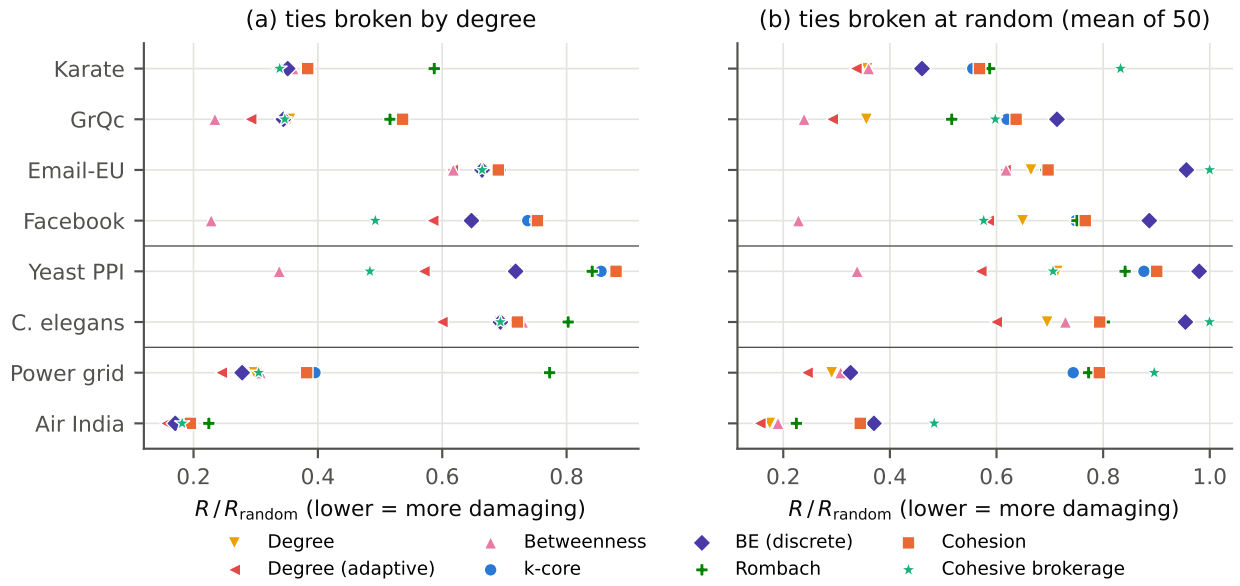


Fig. 7. Robustness relative to random removal, with ties broken (a) by degree and (b) uniformly at random.

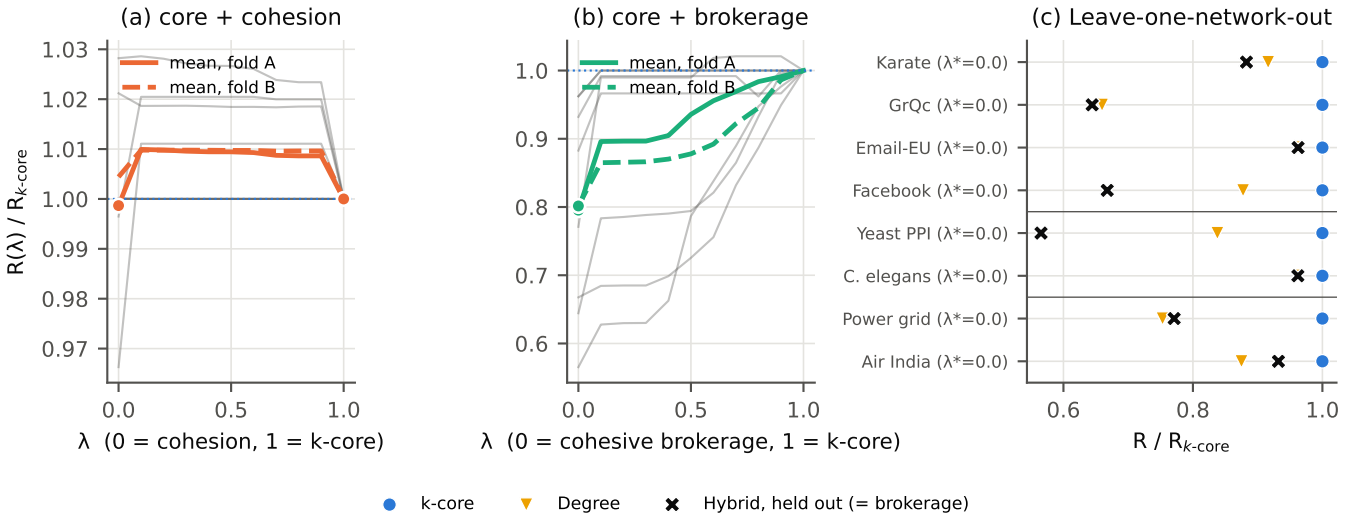


Fig. 8. (a, b) Robustness of the hybrids as a function of λ, relative to k-coreness (grey: individual networks; coloured: training-fold means). (c) Leave-one-network-out evaluation of the core+brokerage hybrid.

dense core, while the cohesion hierarchy is inherently plural.

Implications. Analysts who use k-cores to describe "the core" of a network should check whether it is cohesive; a k-core with $F \gg 1$ is better described as a set of overlapping cohesive groups. For robustness and vulnerability analyses, the useful output of the cohesion hierarchy is not a node's level but its *multiplicity*: brokers that sit in several k-components at once are where the hierarchy can be cut. Cohesive brokerage is parameter-free, reuses the decomposition, and gave its largest gains on the networks where the group-level gap is large—a property that can be measured before any attack is simulated.

Limitations. (i) Eight networks support statements about our sample, not about networks in general; all comparisons are reported with effect sizes, and none survives multiple-testing correction at $\alpha = 0.05$ given the minimum attainable p. (ii) All networks were reduced to simple, undirected, unweighted largest components; weighted or directed cohesion may behave differently. (iii) We study static orderings and the robustness index R; adaptive or optimal-percolation attacks [23] are stronger, and other functional criteria (spreading, synchronisation) may rank the notions differently. Adaptive degree, our strongest baseline, beats brokerage in six networks, and an adaptive variant of brokerage (recomputing the hierarchy after removals) is a natural next step. (iv) For tractability,

Rombach's method used a 6×9 grid instead of 100×100, and BE was optimised greedily from 20 random visiting orders, so both fits may be slightly sub-optimal and the corresponding results are conservative for these methods. (v) Our null model preserves degrees but not clustering; a clustering-preserving null model would test whether the group-level gap exceeds what triangles alone imply. (vi) Exact cohesion remains costly: networks of a few thousand nodes take up to a minute and a half, but very large networks would require approximation or parallelisation.

VIII. Conclusion

Core–periphery cores and structural-cohesion cores agree on who is central but not on how the centre is organised. Across eight networks, k-coreness and cohesion level are practically the same ranking, so cohesion levels neither change the identified core nor improve targeted attacks, and hybrids of the two do not help. The disagreement lives at the group level: in half of the networks one connected k-core is a federation of several overlapping cohesive groups, a feature that is absent from degree-preserving randomisations. The nodes that hold these federations together, the cohesive brokers, are critical: removing them first dismantles the networks markedly faster than coreness, and the gain grows with the size of the group-level gap. We hope that these results, the exact algorithm and the released code help analysts decide when a dense core is also a cohesive one.

Reproducibility

All code, the exact data-cleaning steps, fixed random seeds and the scripts that produce every table and figure are available in the accompanying repository.

References

[1] S. P. Borgatti and M. G. Everett, "Models of core/periphery structures," *Social Networks*, vol. 21, no. 4, pp. 375–395, 2000.

[2] M. P. Rombach, M. A. Porter, J. H. Fowler, and P. J. Mucha, "Core-periphery structure in networks," *SIAM Journal on Applied Mathematics*, vol. 74, no. 1, pp. 167–190, 2014.

[3] S. B. Seidman, "Network structure and minimum degree," *Social Networks*, vol. 5, no. 3, pp. 269–287, 1983.

[4] V. Batagelj and M. Zaveršnik, "An O(m) algorithm for cores decomposition of networks," *arXiv preprint cs/0310049*, 2003.

[5] D. R. White and F. Harary, "The cohesiveness of blocks in social networks: Node connectivity and conditional density," *Sociological Methodology*, vol. 31, no. 1, pp. 305–359, 2001.

[6] J. Moody and D. R. White, "Structural cohesion and embeddedness: A hierarchical concept of social groups," *American Sociological Review*, vol. 68, no. 1, pp. 103–127, 2003.

[7] R. Albert, H. Jeong, and A.-L. Barabási, "Error and attack tolerance of complex networks," *Nature*, vol. 406, no. 6794, pp. 378–382, 2000.

[8] C. M. Schneider, A. A. Moreira, J. S. Andrade, S. Havlin, and H. J. Herrmann, "Mitigation of malicious attacks on networks," *Proceedings of the National Academy of Sciences*, vol. 108, no. 10, pp. 3838–3841, 2011.

[9] P. Holme, "Core-periphery organization of complex networks," *Physical Review E*, vol. 72, no. 4, p. 046111, 2005.

[10] P. Rombach, M. A. Porter, J. H. Fowler, and P. J. Mucha, "Core-periphery structure in networks (revisited)," *SIAM Review*, vol. 59, no. 3, pp. 619–646, 2017.

[11] S. Kojaku and N. Masuda, "Finding multiple core-periphery pairs in networks," *Physical Review E*, vol. 96, no. 5, p. 052313, 2017.

[12] X. Zhang, T. Martin, and M. E. J. Newman, "Identification of core-periphery structure in networks," *Physical Review E*, vol. 91, no. 3, p. 032803, 2015.

[13] R. J. Gallagher, J.-G. Young, and B. Foucault Welles, "A clarified typology of core-periphery structure in networks," *Science Advances*, vol. 7, no. 12, p. eabc9800, 2021.

[14] P. Csermely, A. London, L.-Y. Wu, and B. Uzzi, "Structure and dynamics of core/periphery networks," *Journal of Complex Networks*, vol. 1, no. 2, pp. 93–123, 2013.

[15] M. Kitsak, L. K. Gallos, S. Havlin, F. Liljeros, L. Muchnik, H. E. Stanley, and H. A. Makse, "Identification of influential spreaders in complex networks," *Nature Physics*, vol. 6, no. 11, pp. 888–893, 2010.

[16] Y. Liu, M. Tang, T. Zhou, and Y. Do, "Core-like groups result in invalidation of identifying super-spreader by k-shell decomposition," *Scientific Reports*, vol. 5, p. 9602, 2015.

[17] K. Burleson-Lesser, F. Morone, M. S. Tomassone, and H. A. Makse, "K-core robustness in ecological and financial networks," *Scientific Reports*, vol. 10, p. 3357, 2020.

[18] J. Torrents and F. Ferraro, "Structural cohesion: Visualization and heuristics for fast computation," *Journal of Social Structure*, vol. 16, no. 8, 2015.

[19] A. A. Hagberg, D. A. Schult, and P. J. Swart, "Exploring network structure, dynamics, and function using NetworkX," in *Proceedings of the 7th Python in Science Conference (SciPy2008)*, 2008, pp. 11–15.

[20] S. Even, "An algorithm for determining whether the connectivity of a graph is at least k," *SIAM Journal on Computing*, vol. 4, no. 3, pp. 393–396, 1975.

[21] A.-H. Esfahanian, "Connectivity algorithms," in *Topics in Structural Graph Theory*, L. W. Beineke and R. J. Wilson, Eds. Cambridge University Press, 2013, pp. 268–281.

[22] T. Łuczak, "Size and connectivity of the k-core of a random graph," *Discrete Mathematics*, vol. 91, no. 1, pp. 61–68, 1991.

[23] F. Morone and H. A. Makse, "Influence maximization in complex networks through optimal percolation," *Nature*, vol. 524, no. 7563, pp. 65–68, 2015.

[24] K. Menger, "Zur allgemeinen Kurventheorie," *Fundamenta Mathematicae*, vol. 10, pp. 96–115, 1927.

[25] S. Kirkpatrick, C. D. Gelatt, and M. P. Vecchi, "Optimization by simulated annealing," *Science*, vol. 220, no. 4598, pp. 671–680, 1983.

[26] J. Hopcroft and R. Tarjan, "Algorithm 447: Efficient algorithms for graph manipulation," *Communications of the ACM*, vol. 16, no. 6, pp. 372–378, 1973.

[27] S. K. Lam, A. Pitrou, and S. Seibert, "Numba: A LLVM-based Python JIT compiler," in *Proceedings of the Second Workshop on the LLVM Compiler Infrastructure in HPC*, 2015, pp. 1–6.

[28] N. X. Vinh, J. Epps, and J. Bailey, "Information theoretic measures for clusterings comparison: Variants, properties, normalization and correction for chance," *Journal of Machine Learning Research*, vol. 11, pp. 2837–2854, 2010.

[29] S. Maslov and K. Sneppen, "Specificity and stability in topology of protein networks," *Science*, vol. 296, no. 5569, pp. 910–913, 2002.

[30] M. E. J. Newman and R. M. Ziff, "Efficient Monte Carlo algorithm and high-precision results for percolation," *Physical Review Letters*, vol. 85, no. 19, pp. 4104–4107, 2000.

[31] U. Brandes, "A faster algorithm for betweenness centrality," *Journal of Mathematical Sociology*, vol. 25, no. 2, pp. 163–177, 2001.

[32] B. Efron, "Bootstrap methods: Another look at the jackknife," *The Annals of Statistics*, vol. 7, no. 1, pp. 1–26, 1979.

[33] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bulletin*, vol. 1, no. 6, pp. 80–83, 1945.

[34] D. S. Kerby, "The simple difference formula: An approach to teaching nonparametric correlation," *Comprehensive Psychology*, vol. 3, p. 11.IT.3.1, 2014.

[35] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.

[36] W. W. Zachary, "An information flow model for conflict and fission in small groups," *Journal of Anthropological Research*, vol. 33, no. 4, pp. 452–473, 1977.

[37] J. Leskovec, J. Kleinberg, and C. Faloutsos, "Graph evolution: Densification and shrinking diameters," *ACM Transactions on Knowledge Discovery from Data*, vol. 1, no. 1, p. 2, 2007.

[38] H. Yin, A. R. Benson, J. Leskovec, and D. F. Gleich, "Local higher-order graph clustering," in *Proceedings of the 23rd ACM SIGKDD*

International Conference on Knowledge Discovery and Data Mining, 2017, pp. 555–564.

[39] J. McAuley and J. Leskovec, "Learning to discover social circles in ego networks," in *Advances in Neural Information Processing Systems 25*, 2012, pp. 539–547.

[40] J. Leskovec and A. Krevl, "SNAP Datasets: Stanford large network dataset collection," http://snap.stanford.edu/data, 2014.

[41] S. R. Collins, P. Kemmeren, X.-C. Zhao, J. F. Greenblatt, F. Spencer, F. C. P. Holstege, J. S. Weissman, and N. J. Krogan, "Toward a comprehensive atlas of the physical interactome of *Saccharomyces cerevisiae*," *Molecular & Cellular Proteomics*, vol. 6, no. 3, pp. 439–450, 2007.

[42] J. G. White, E. Southgate, J. N. Thomson, and S. Brenner, "The structure of the nervous system of the nematode *Caenorhabditis elegans*," *Philosophical Transactions of the Royal Society of London B*, vol. 314, no. 1165, pp. 1–340, 1986.

[43] D. J. Watts and S. H. Strogatz, "Collective dynamics of 'small-world' networks," *Nature*, vol. 393, no. 6684, pp. 440–442, 1998.

[44] T. P. Peixoto, "The Netzschleuder network catalogue and repository," https://networks.skewed.de, 2020.

[45] OpenFlights, "Airport, airline and route data," https://openflights.org/data.html, accessed October 2026.

[46] R. S. Burt, *Structural Holes: The Social Structure of Competition*. Cambridge, MA: Harvard University Press, 1992.